

CS & IT ENGINEERING



Computer Network - 1

Transport Layer

Lecture No. – 01

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Recap of Previous Lecture



Topic

Distance Vector Routing

Topic

Routing Protocols



Topics to be Covered



Topic

Transport Layer

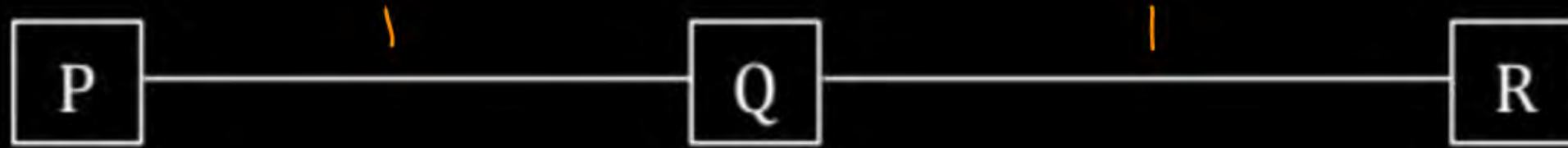


[NAT]

[GATE-2022][1 Mark]

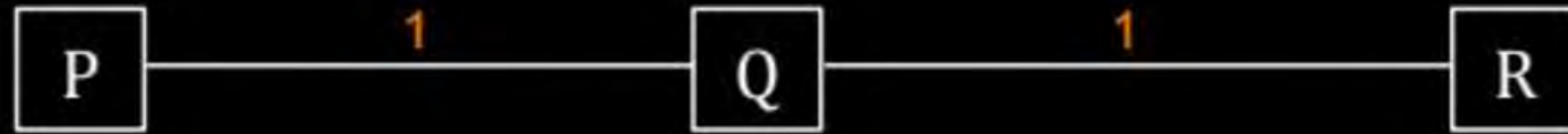


#Q. Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity.



The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q-R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (rounded off to one decimal place) between P and Q, leading to count-to-infinity problem, is _____.

$$\text{Ans} = \frac{1}{2} = 0.5$$

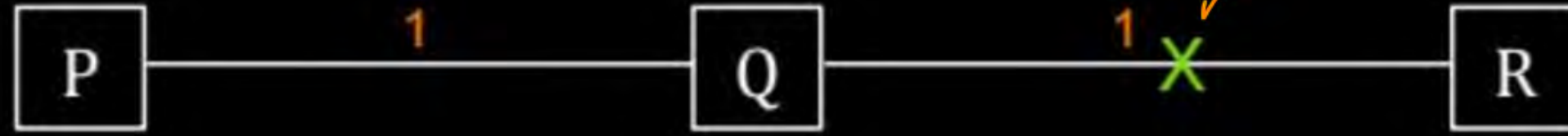


Node	Cost	Via
P	0	P
Q	1	Q
R	2	Q

Node	Cost	Via
P	1	P
Q	0	Q
R	1	R

Node	Cost	Via
P	2	Q
Q	1	Q
R	0	R

CASE I :-



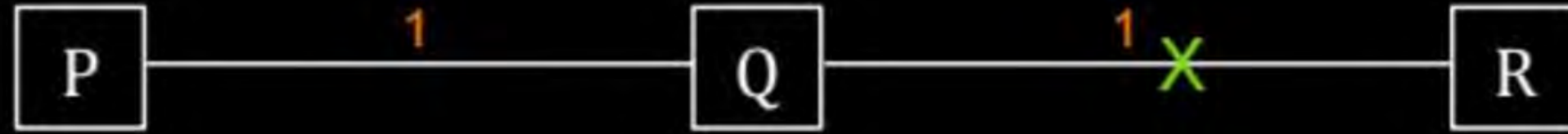
Node	Cost	Via
P	0	P
Q	1	Q
R	2	Q

Node	Cost	Via
P	1	P
Q	0	Q
R	∞	

Node	Cost	Via
P	∞	
Q	∞	
R	0	R

P	1
Q	0
R	∞

CASE II :-



Node	Cost	Via
P	0	P
Q	1	Q
R	2	Q

Node	Cost	Via
P	1	P
Q	0	Q
R	∞	

Node	Cost	Via
P	∞	
Q	∞	
R	0	R

P	0
Q	1
R	2

P	1
Q	0
R	∞

[MCQ]

[GATE-IT-2008] [2 Mark]



#Q. Two popular routing algorithms are Distance Vector (DV) and Link State (LS) routing. Which of the following are true?

- ✓ (S1) Count to infinity is a problem only with DV and not LS routing T
- (S2) In LS, the shortest path algorithm is run only at one node F
- (S3) In DV, the shortest path algorithm is run only at one node F
- ✓ (S4) DV requires lesser number of network messages than LS T

A S1, S2 and S4 only

B S1, S3 and S4 only

C S2 and S3 only

✓ D S1 and S4 only

Ans: D

#Q. Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links.

[S1] The computational overhead in link state protocols is higher than in distance vector protocols. TRUE

[S2] A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol. FALSE

[S3] After a topology change, a link state protocol will converge faster than a distance vector protocol. TRUE

Which one of the following is correct about S1, S2, and S3 ?

Ans: D

- A S1, S2, and S3 are all true
- B S1, S2, and S3 are all false
- C S1 and S2 are true, but S3 is false
- ✓ D S1 and S3 are true, but S2 is false

[MCQ]

[GATE-2011][2 Mark]



#Q. Consider a network with five nodes, N1 to N5, as shown below. The network uses a Distance Vector Routing protocol. (Once the routes have stabilized) the distance vectors at different nodes are as following.

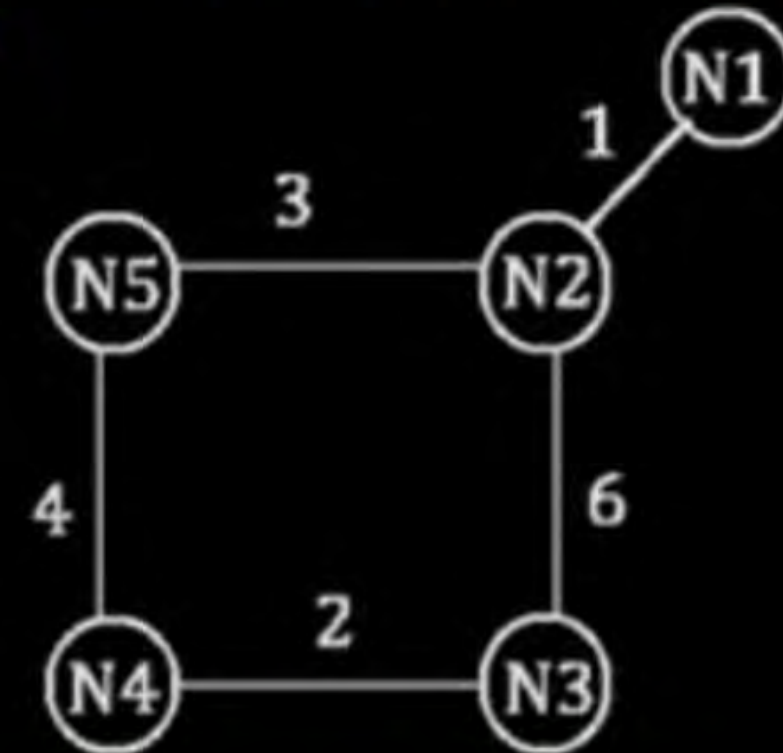
N1 : (0, 1, 7, 8, 4)

N2 : (1, 0, 6, 7, 3)

N3 : (7, 6, 0, 2, 6)

N4 : (8, 7, 2, 0, 4)

N5 : (4, 3, 6, 4, 0)



Each distance vector is the distance of the best known path at that instance to nodes, N1 to N5, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors.

In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

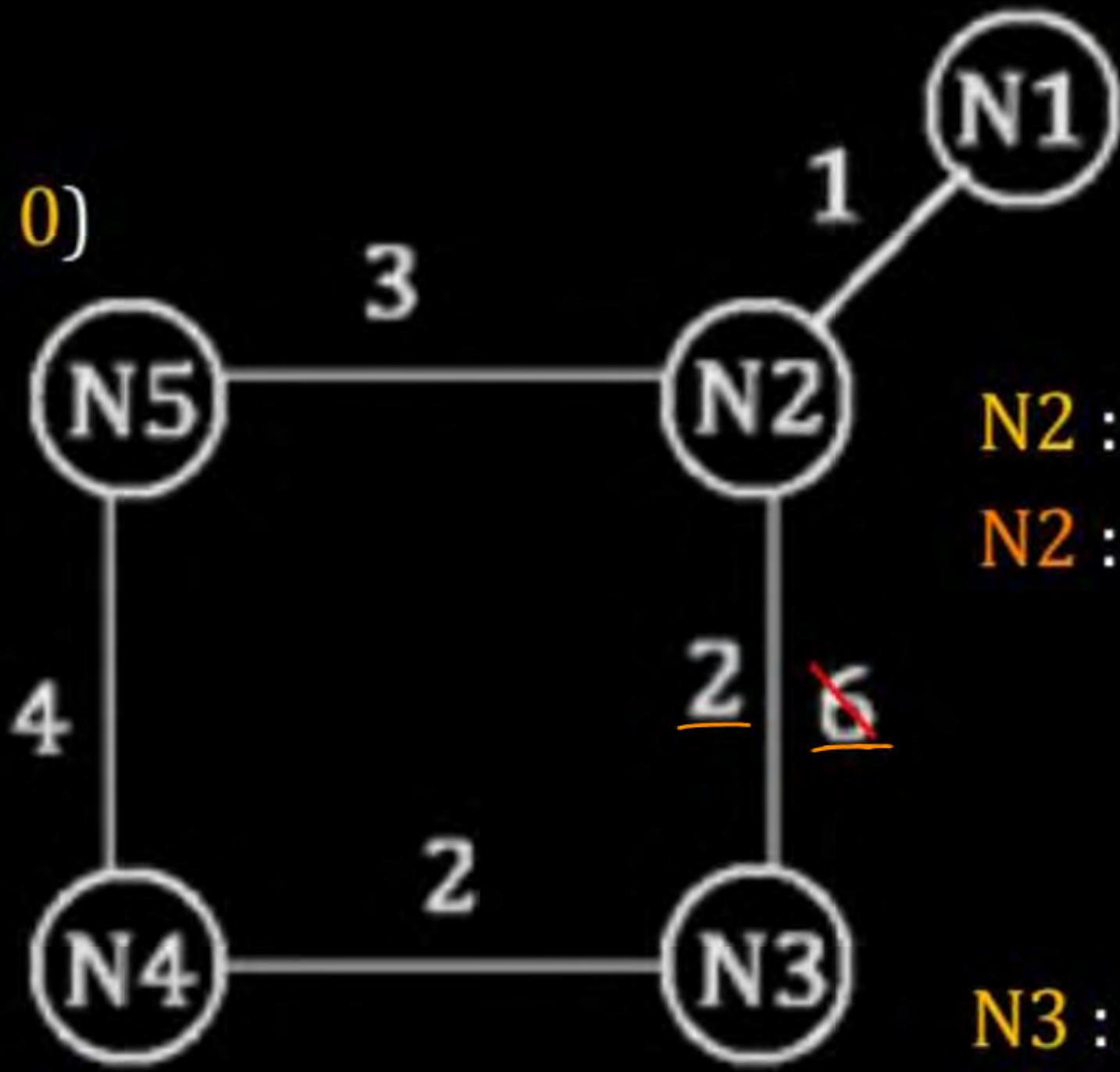
The cost of link N2-N3 reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, N3?

- ☒ A (3, 2, 0, 2, 5)
- ☐ B (3, 2, 0, 2, 6)
- ☐ C (7, 2, 0, 2, 5)
- ☐ D (7, 2, 0, 2, 6)

Ans: A

N1 : (0, 1, 7, 8, 4)

N5 : (4, 3, 6, 4, 0)



N2 : (1, 0, 6, 7, 3)

N2 : (1, 0, 2, 7, 3)

N3 : (7, 6, 0, 2, 6)

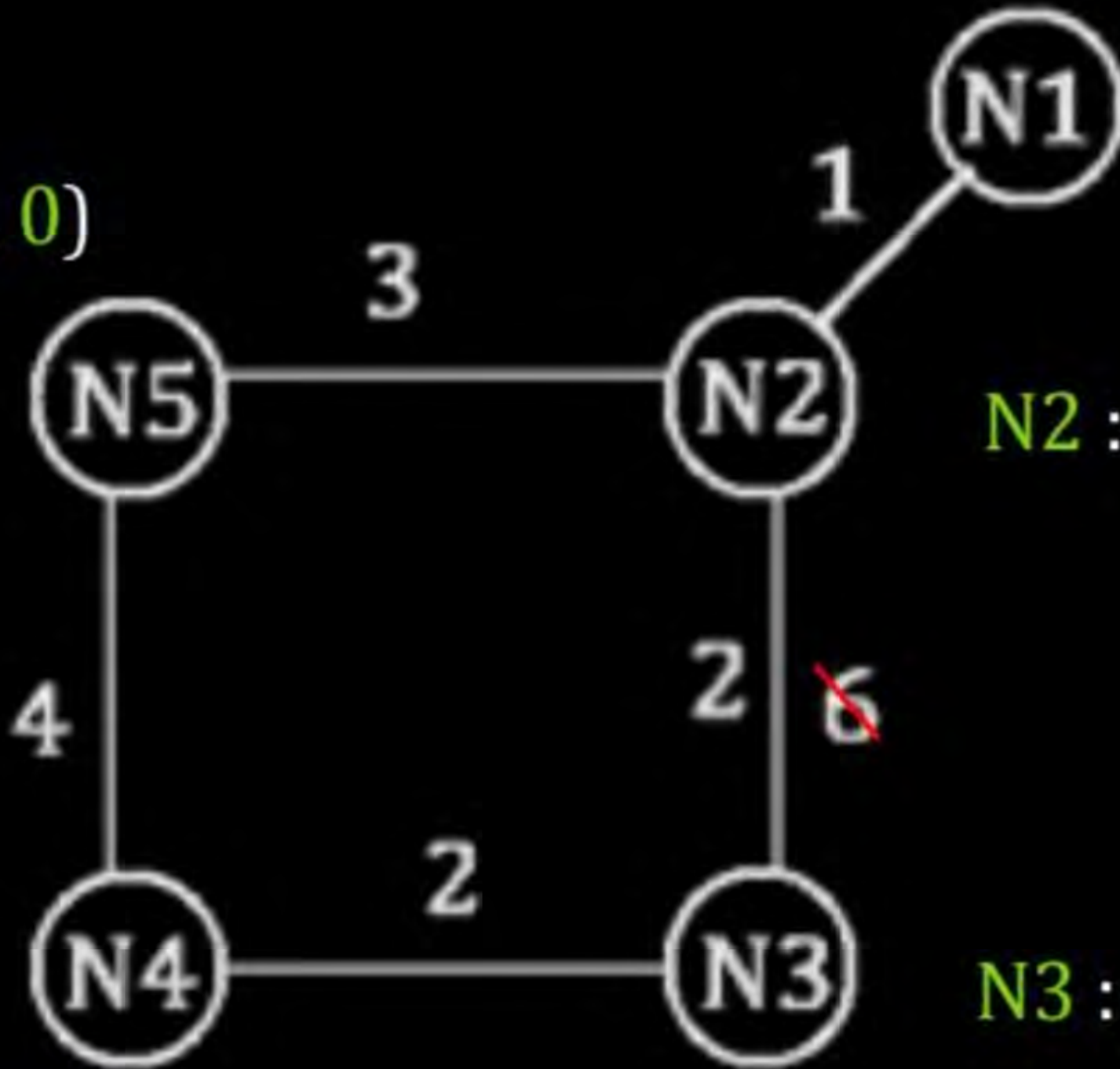
N3 : (7, 2, 0, 2, 6)

N4 : (8, 7, 2, 0, 4)

N4

N1 : (0, 1, 3, 8, 4)

N5 : (4, 3, 5, 4, 0)



N2 : (1, 0, 2, 4, 3)
N₁

N3 : (3, 2, 0, 2, 5)
N₂ N₂ N₄ N₂

N4 : (8, 4, 2, 0, 4)
N₅

[MCQ]

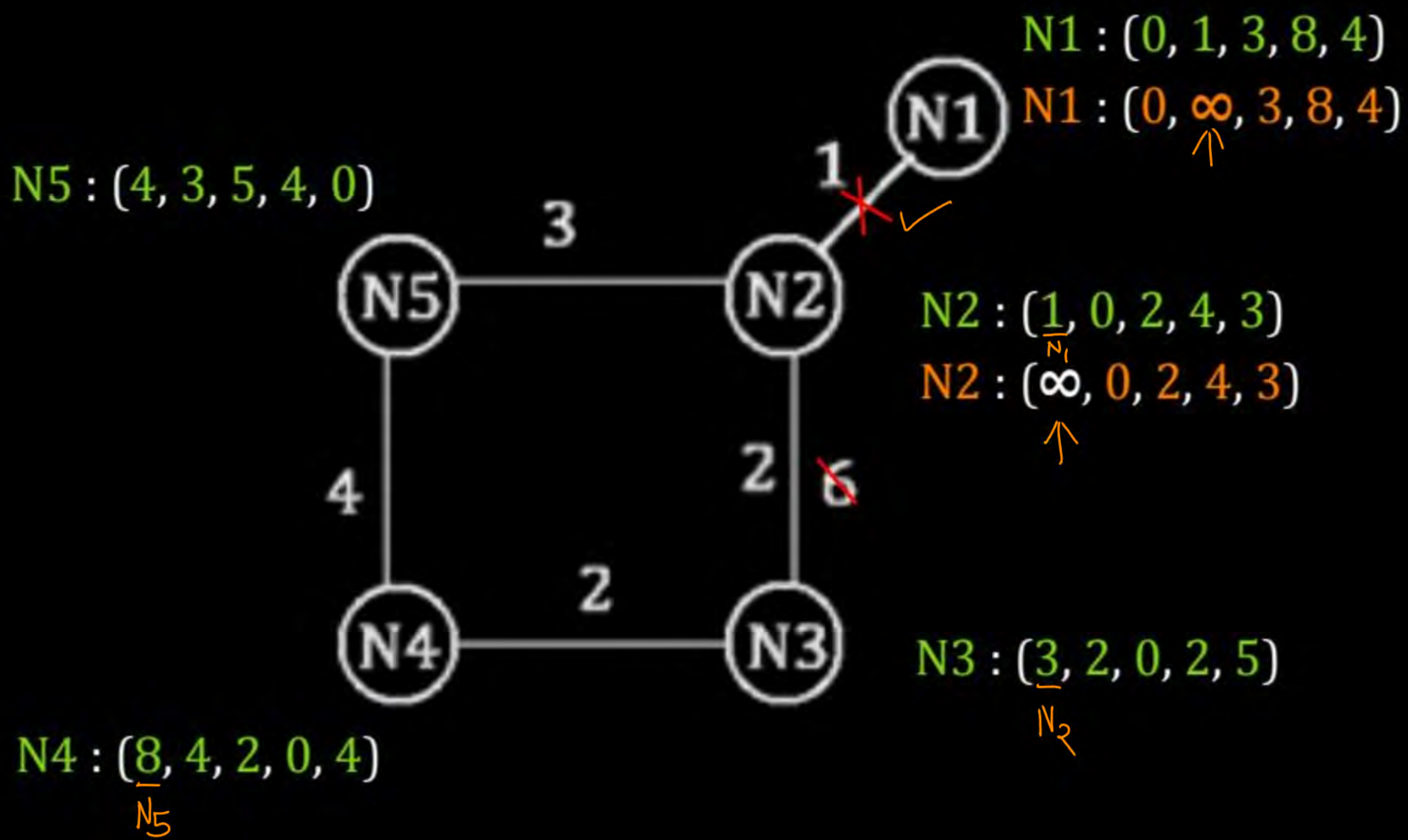
[GATE-2011][2 Mark]



#Q. Consider the same data as given in previous question. After the update in the previous question, the link N1-N2 goes down. N2 will reflect this change immediately in its distance vector as cost, infinite. After the NEXT ROUND of update, what will be cost to N1 in the distance vector of N3 ?

- ☐ A 3
- ☐ B 9
- ☒ C 10
- ☐ D Infinite

Ans: C



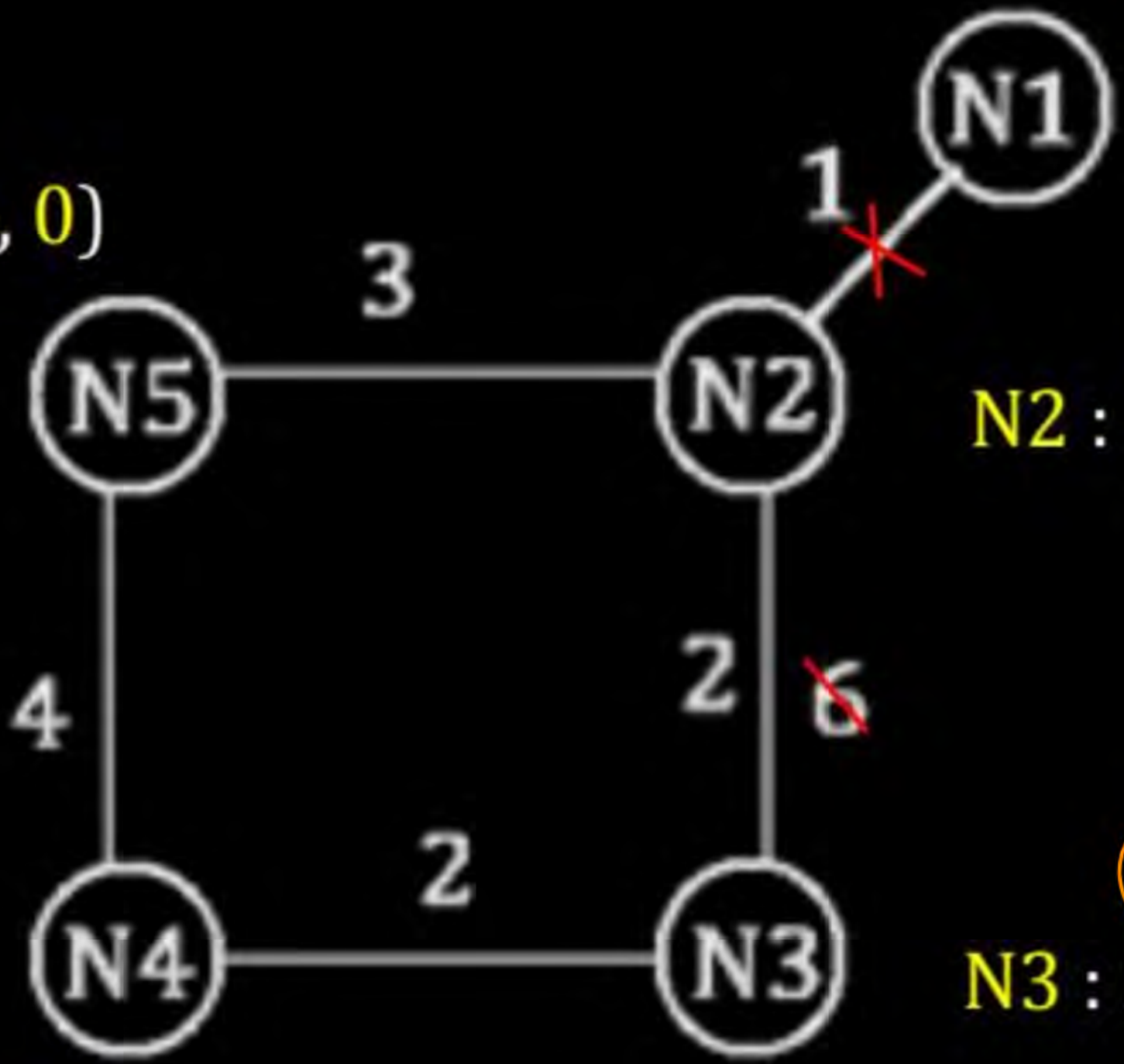
N1 : (0, ∞, ∞, ∞, ∞)

↓
N5 : (∞, 3, 5, 4, 0)

N2 : (∞, 0, 2, 4, 3)

N3 : (10, 2, 0, 2, 5)
N4

N4 : (5, 4, 2, 0, 4)



[MCQ]

[GATE-2017] [1 Mark]



#Q. Consider the following statements about the routing protocols, Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.

- ✓ I. RIP uses distance vector routing (RIP-DV) True
- ✓ II. RIP packets are sent using UDP (RIP → UDP → IP) True
- III. OSPF packets are sent using TCP (OSPF → IP) False
- ✓ IV. OSPF operation is based on link-state routing (OSPF-LS) True

Which of the following above are CORRECT?

- A I and IV only
- B I, II and III only
- ✓ C I, II and IV only
- D II, III and IV only

Ans: C

[MSQ]

[GATE-2023][1 Mark]



#Q. Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?

☒ A OSPF implements Bellman-Ford algorithm to find shortest paths. False

☐ B OSPF uses Dijkstra's shortest path algorithm to implement least-cost path routing. TRUE

☒ C OSPF is used as an inter-domain routing protocol. FALSE
↳ (Intra-domain) ↳ BGP

☐ D OSPF implements hierarchical routing. TRUE

Ans: A & C

[MCQ]

[GATE-2020] [1 Mark]



#Q. Consider the following statements about the functionality of an IP based router.

- I. A router does not modify the IP packets during forwarding. FALSE
- II. It is not necessary for a router to implement any routing protocol. TRUE
- III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet. FALSE

Which of the above statements is/are TRUE?

- A I and II only
- B I only
- C II and III only
- D ☒ II only

Ans: D



Topic : Syllabus



Transport layer : flow control and congestion control, UDP, TCP, sockets;

↓
TCP



Topic : Transport Layer



→ Provide logical communication between application processes

(Processes running on different machine)

→ Responsible for process-to-process (end-to-end) communication (Connectivity)

→ Multiplexing & Demultiplexing

→ Not Host-to-Host

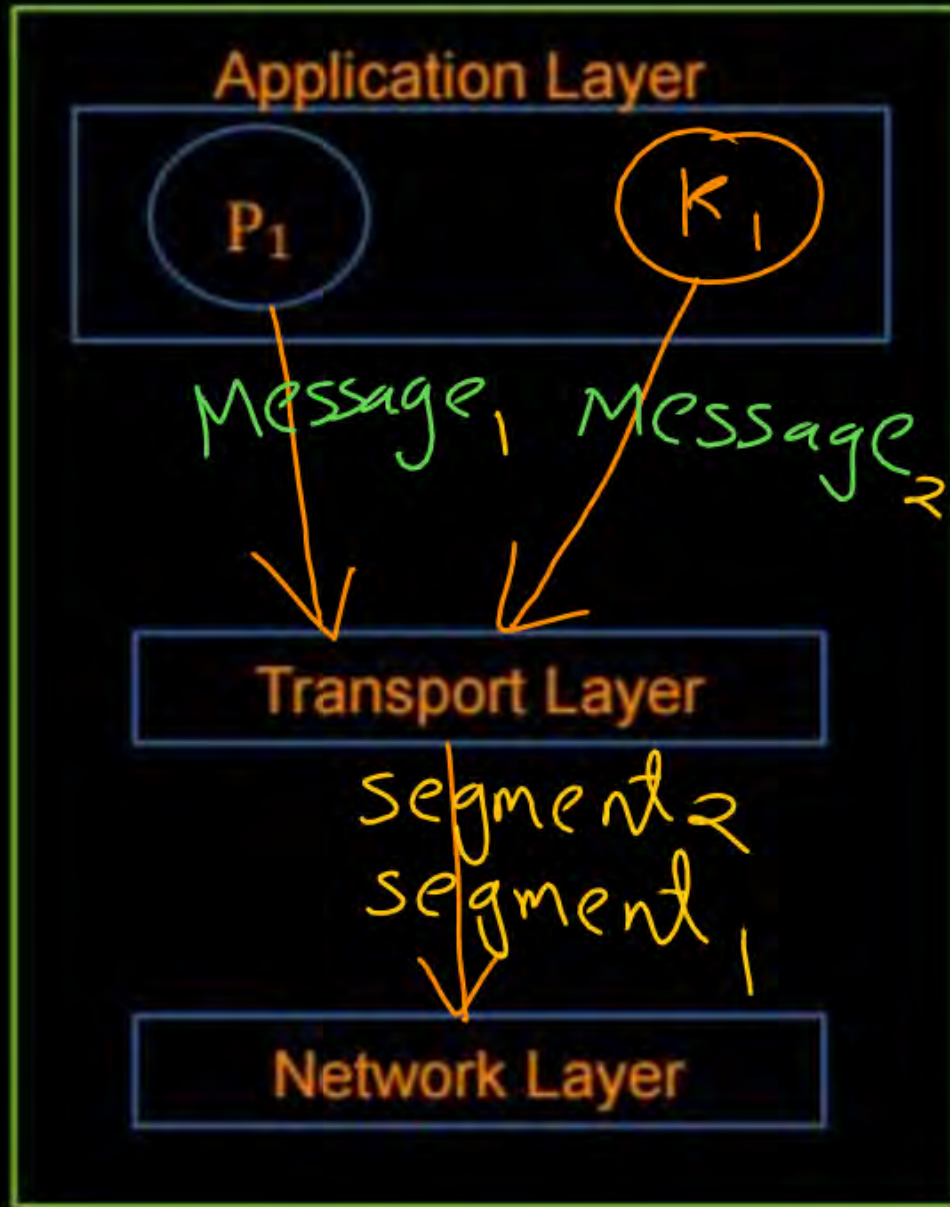
→ Socket-to-Socket



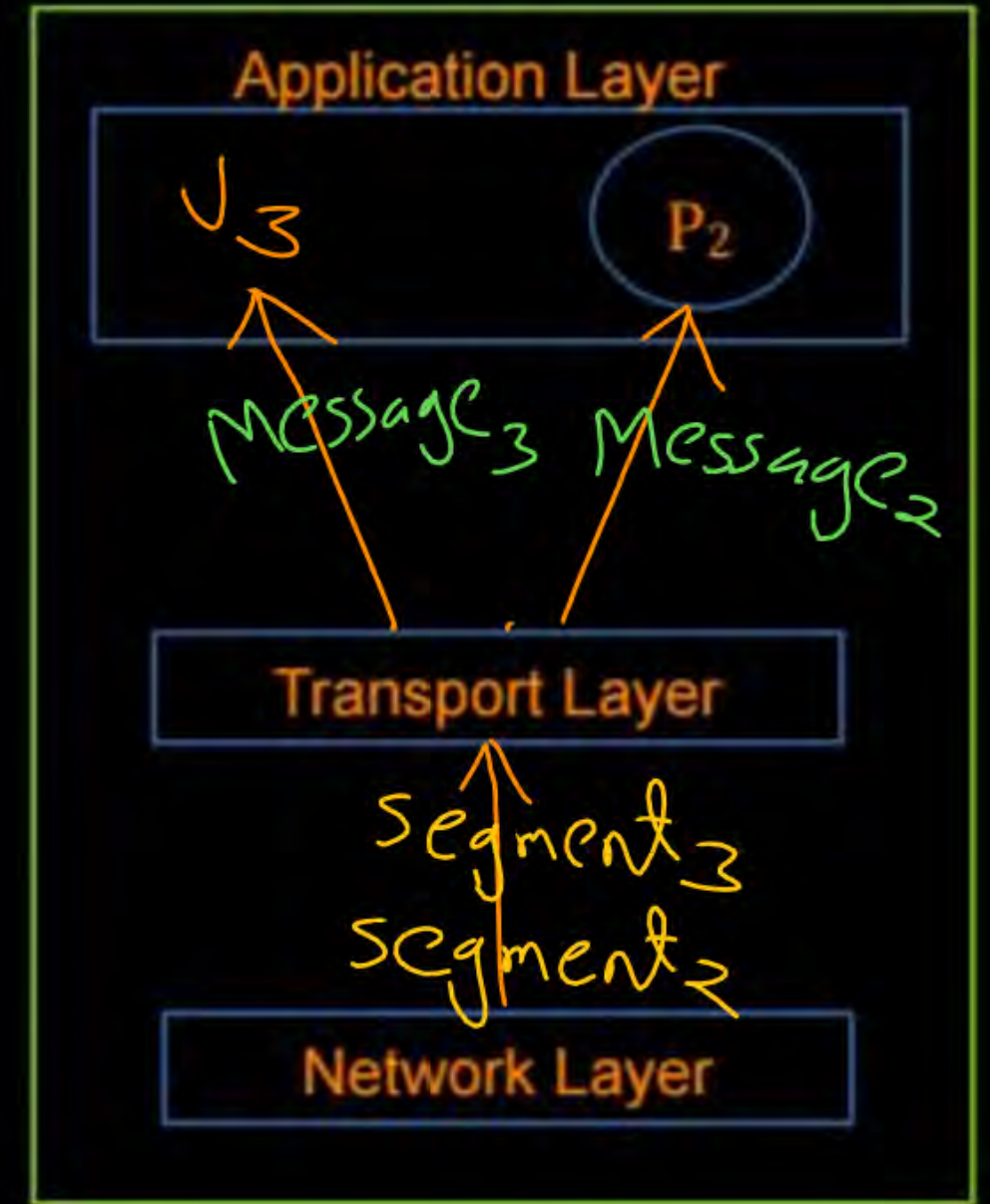
Topic : Transport Layer



Source Host



Destination Host





Topic : Socket



- One endpoint of a two-way communication
- Two network process communicate via a pair of sockets
- Door or interface of a network process
- Data structure



Topic : Multiplexing



Multiplexing at sender :

- > Handle data from multiple sockets
- > Add transport layer protocol header
[Used for demultiplexing at receiver]

Source Port No.
Dest. Port No.



Topic : Transport Layer

- > Transport Layer PDU : "Segment"
- > Sender : Divide application ^(DATA) messages into segments,
Segments passes to network layer
- > Receiver : Reassemble segments into messages,
Messages passes to application layer



Topic : Transport Layer



Two Transport Layer Protocols :

1. UDP : User Datagram Protocol
2. TCP : Transmission Control Protocol

UDP : connection-less (stateless)

TCP : Connection-Oriented (stateful)



Topic : Transport Layer



Application Layer Protocol	Transport Layer Protocol
<u>DNS</u>	<u>UDP</u> (default), <u>TCP</u> (conditional)
<u>HTTP 1 & HTTP 2</u>	<u>TCP</u>
<u>HTTP 3</u>	<u>UDP</u>
<u>FTP</u> , <u>SMTP</u> , <u>IMAP</u> (<u>POP</u>)	<u>TCP</u>
<u>DHCP</u>	<u>UDP</u>
<u>RIP</u>	<u>UDP</u>
<u>BGP</u>	<u>TCP</u>



Topic : UDP Socket



UDP $\rightarrow$ Datagram Oriented



Topic : TCP Socket



TCP $\rightarrow$ stream oriented
(byte-stream)



Topic : Socket



=> Socket is uniquely identified by a combination of three factors :

- IP address → Identification of Host
- Port number → Identification of process within host
- Transport protocol ⇒ TCP socket
or
UDP socket



2 mins Summary



Topic

Transport Layer Services → Socket

Topic

~~UDP~~



THANK - YOU

